

9.15 Solution: (a) From Prob. 6.21, we can see that the lowest non-vanishing moments are magnetic dipole and electric quadrupole. For a single dipole, its magnetic moment is

$$\frac{1}{2}\mathbf{p} \times \mathbf{v} = \frac{1}{2}\mathbf{p} \times (\boldsymbol{\omega} \times \mathbf{r}) = -\frac{1}{2}\omega pa \hat{\mathbf{r}} = -\frac{1}{2}\omega pa (\hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi).$$

Therefore, for the two-dipole configuration as described in the problem, the magnetic dipole is

$$\begin{aligned} \mathbf{m}(t) &= -\frac{1}{2}\omega pa [(\hat{\mathbf{x}} \cos(\omega t) + \hat{\mathbf{y}} \sin(\omega t)) - (\hat{\mathbf{x}} \cos(\omega t + \pi) + \hat{\mathbf{y}} \sin(\omega t + \pi))] \\ &= -\omega pa (\hat{\mathbf{x}} \cos(\omega t) + \hat{\mathbf{y}} \sin(\omega t)) = \text{Re} [\omega pa(-1, -i, 0)e^{-i\omega t}]. \end{aligned}$$

For the quadrupole moment, using the result of Prob. 4.2, we can write the charge density of the two-particle configuration as

$$\rho(\mathbf{x}) = -(\mathbf{p} \cdot \nabla) \left[\frac{1}{r^2} \delta(r-a) \delta(\cos \theta) (\delta(\phi - \omega t) - \delta(\phi - \omega t - \pi)) \right],$$

and the quadrupole moment can be calculated as

$$\begin{aligned} Q_{\alpha\beta}(t) &= \int (3x_\alpha x_\beta - r^2 \delta_{\alpha\beta}) \rho(\mathbf{x}) d^3x \\ &= \int \frac{1}{r^2} \delta(r-a) \delta(\cos \theta) [\delta(\phi - \omega t) - \delta(\phi - \omega t - \pi)] \left(p \frac{\partial}{\partial z} \right) (3x_\alpha x_\beta - r^2 \delta_{\alpha\beta}) d^3x. \end{aligned}$$

It is straightforward to verify that only Q_{xz} and Q_{yz} are non-zero, which are given by

$$\begin{aligned} Q_{xz}(t) &= \int \frac{1}{r^2} \delta(r-a) \delta(\cos \theta) [\delta(\phi - \omega t) - \delta(\phi - \omega t - \pi)] 3px d^3x \\ &= 3pa [\cos(\omega t) - \cos(\omega t + \pi)] = 6pa \cos(\omega t) \\ &= \text{Re} [6pae^{-i\omega t}], \end{aligned}$$

and

$$Q_{yz}(t) = 6pa \sin(\omega t) = \text{Re} [6ipae^{-i\omega t}].$$

(b) With both magnetic dipole and electric quadrupole moments present, the radiating magnetic field is, as given by Eqs. (9.35) and (9.44),

$$\mathbf{H} = -\frac{k^2}{4\pi} \mathbf{n} \times \left[\mathbf{n} \times \mathbf{m} + \frac{1\omega}{6} \mathbf{Q}(\mathbf{n}) \right] \frac{e^{ikr}}{r},$$

where

$$\mathbf{Q}(\mathbf{n}) = \sum_{\beta} Q_{\alpha\beta} n_{\beta} = 6pa \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & i \\ 1 & i & 0 \end{pmatrix} \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} = 6pa \begin{pmatrix} \cos \theta \\ i \cos \theta \\ \sin \theta e^{i\phi} \end{pmatrix}.$$

Also,

$$\mathbf{n} \times \mathbf{m} = \omega pa \left(\hat{\mathbf{x}}(i \cos \theta) + \hat{\mathbf{y}}(-\cos \theta) + \hat{\mathbf{z}}(-i \sin \theta e^{i\phi}) \right).$$

Thus,

$$\mathbf{n} \times \mathbf{m} + \frac{1\omega}{6} \mathbf{Q}(\mathbf{n}) = \omega pa \left[\left(\hat{\mathbf{x}}(i \cos \theta) + \hat{\mathbf{y}}(-\cos \theta) + \hat{\mathbf{z}}(-i \sin \theta e^{i\phi}) \right) \right]$$

$$\begin{aligned}
& + \left(\hat{\mathbf{x}}(i \cos \theta) + \hat{\mathbf{y}}(-\cos \theta) + \hat{\mathbf{z}}(i \sin \theta e^{i\phi}) \right) \Big] \\
& = 2\omega p a [\hat{\mathbf{x}}(i \cos \theta) + \hat{\mathbf{y}}(-\cos \theta)],
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{H} &= -\frac{k^2}{4\pi} \mathbf{n} \times \left[\mathbf{n} \times \mathbf{m} + \frac{i\omega}{6} \mathbf{Q}(\mathbf{n}) \right] \frac{e^{ikr}}{r} \\
&= -\frac{cpa}{2\pi} k^3 \mathbf{n} \times [\hat{\mathbf{x}}(i \cos \theta) + \hat{\mathbf{y}}(-\cos \theta)] \frac{e^{ikr}}{r} \\
&= -\frac{cpa}{2\pi} k^3 \left[(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) \cos \theta - \hat{\mathbf{z}} \sin \theta e^{i\phi} \right] \cos \theta \frac{e^{ikr}}{r},
\end{aligned}$$

which agrees with Jackson up to a sign.

(c) From Eq. (9.21),

$$\frac{dP}{d\Omega} = \frac{Z_0}{2} r^2 |\mathbf{H}|^2 = \frac{Z_0}{2} \frac{c^2 p^2 a^2}{4\pi^2} k^6 (2 \cos^2 \theta + \sin^2 \theta) \cos^2 \theta = \frac{1}{8\pi^2 \varepsilon_0} c k^6 p^2 a^2 (\cos^2 \theta + \cos^4 \theta).$$

The total time averaged power radiated is

$$P = \int \frac{dP}{d\Omega} d\Omega = \frac{1}{8\pi^2 \varepsilon_0} c k^6 p^2 a^2 2\pi \int_{-1}^1 (x^2 + x^4) dx = \frac{1}{4\pi \varepsilon_0} c k^6 p^2 a^2 \frac{16}{15} = \frac{4}{15\pi \varepsilon_0} c k^6 p^2 a^2.$$